

LAERDON KIM

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EDUCATION

Cornell University — GPA: 4.14/4.0

B.A., Computer Science

Ithaca, NY

Expected May 2028

- Activities: Association of CS Undergraduates Academic Lead, Cornell Undergraduate AI Research. INFO 1998 Lecturer. CLPsych PC. Machine Learning TA. **Coursework:** Advanced Language Technologies - *Grad* (A+), Natural Language Processing (A+), Machine Learning (A+)
- Awards: Milstein Scholar, International Mathematical Contest in Modeling (Qualifier), IMC Prosperity Top 250 (~1.3% of teams), Oxford Cosmos Institute Grantee (~8% acceptance), 1st place, MIT AI + Education Hackathon

EXPERIENCE

Researcher — Cornell University, NLP Group

September 2024 – Present

- Researching novel LLM training schemes for forecasting online toxicity in time-series conversational data; designed the first decision deferral system for forecasting (ACL Main 2026).
- Building simulation and eval infrastructure to detect misalignment and audit neutrality in AI civic discourse systems (Laude Moonshot).
- Launched large-scale AI moderation study on Wikipedia supported by the Wikimedia Foundation's first multi-year grant.
- Researching the formation of word concepts in language model activations; researching data + pretraining to combat anisotropy. Co-authored EMNLP Main 2025 paper exposing critical flaw in encoder models on semantic change detection tasks, proposing new eval.
- Solo-authored publication in NAACL 2025, CLPsych Workshop. Created evaluation pipeline for LLM identification of self-state spans in mental health data. **Won third out of fourteen systems.** Automated training workflows with Bash scripting on Linux HPC clusters.

Contributor — ConvoKit (Open-source NLP Package)

September 2024 – Present

- 1 of 5 primary maintainers. Authored v4.1.2, developing DecisionPolicy class for learned intervention systems; contributor v3.0.1.
- Reviewed and merged PRs for bug fixes, added 4 new datasets, and wrote 6 demo Jupyter notebooks. Modernized backend for forecasting system with Docker containerization. ~30k lines contributed. Engineered CI/CD pipelines to automate deployment.
- Developed modular data loaders and preprocessing scripts to handle LLM inference on large dialogue datasets (10k+ conversations).

Visiting Researcher — Illinois Institute of Technology

June 2024 – August 2024

- Coded custom CUDA fused attention and LayerNorm kernels for a 1.2x inference speedup on diffusion models.
- Utilized z-score and absolute value threshold outlier masking methods on LLaMA 2, resulting in a 6% reduction in perplexity and 13% reduction in memory usage. Re-implemented SmoothQuant and LLM-AWQ on OPT-8B for text completion inference.

PUBLICATIONS & PRESENTATIONS

Laerdon Kim, Vivian Nguyen, Cristian Danescu-Niculescu-Mizil. *Wait! There's a Way Out: A Decision Mechanism for Forecasting Conversational Derailment*. ACL 2026 Main. Top 2.8% of submissions.

Khonzoda Umarova, Lillian Lee, Laerdon Kim. *Current Semantic-change Quantification Methods Struggle with Semantic-change Discovery in the Wild*. EMNLP 2025 Main.

Laerdon Kim. *A Baseline for Self-state Identification and Classification in Mental Health Data: CLPsych 2025 Task*. Accepted at NAACL 10th Workshop on Computational Linguistics and Clinical Psychology (CLPsych). arXiv:2504.14066

Laerdon Kim, Jenny Jin, Ajahla Jefferson, Tina Zeng, Brian Chica-Herrera. "Social Mobility in the Mission Hill/Roxbury Area." Presentation at Opportunity Insights Lab, Harvard University, Boston, MA, May 2024.

PROJECTS

Byte-ing Off More Than You Can Chew | Byte-level transformer pretraining

2026

- Reproduced and extended Meta's Byte Latent Transformer, discovering parameter balances superior to the original at ultra-small (86M) scale: through ablations across 6 architectural variants, discovered optimality of context embedding map consolidation.
- Implemented a novel sinusoidal patch-length embedding (added to patch states before the global transformer), improving perplexity. Built reproducible distributed training infrastructure from scratch on Prime Intellect H100s. Funniest Poster award.

Git Distributed | Distributed systems Git clone

2025

- Implemented multithreaded primary-backup backend in C++ for a Git-like CLI for concurrent pushes and pulls, adding health checks and automatic backup promotion to maintain availability under primary failure.

Vocera | Application and custom encoder for deepfake detection

2025

- Designed and trained composite encoder using wav2vec embeddings, feature extraction with OpenSMILE, and speech-to-text with OpenAI Whisper. Built React Native app at UC Berkeley AI Hackathon for real-time voice verification.

Languages: Python (proficient), Java (proficient), Swift (proficient), C++ (intermediate). **Tools:** PyTorch, W&B, Linux, Docker